

The resolver that remembered

Architecting the Agentic Enterprise: Middleware, Durable State, and Event-Driven AI

Investigate • Pause • Recover • Settle once

03

45
MINUTES

The 45-minute route

Time	Activity	Evidence
0-6	Read the dispute; declare mode	Case facts and mode
6-13	Assign state owners	Workflow + payment identities
13-23	Investigate and pause	Graph reads + validated proposal
23-35	Crash and redeliver	Same pending workflow
35-41	Approve and inspect	Exactly one settlement
41-45	Review limits	Durability responsibilities

Follow the worksheet; record the actual evidence from your run.

The customer leaves. The work continues.

CASE

case-501 reports a disputed charge. Investigate before deciding entitlement.

PROPOSAL

The resolver gathers governed evidence and proposes a resolution.

OUTCOME

Approval leads to one settlement; rejection creates no settlement.

The business process must outlive both the chat and the worker.

Declare how the investigation is running

LIVE

Compiled LangGraph calls the configured model through Gate 1 and selects allowed reads.

OFFLINE REPLAY

Fixtures drive the same graph; governed reads and banking controls still execute.

HONEST LABELS

A provider failure remains a failure. Recorded text is not newly verified evidence.

Replay case-501 proposes 75,000 paise / ₹750 to acc-101.

Give every kind of state an owner

Component	Owns	Recovery role
Kafka	Case-event delivery	Redelivery is expected
Temporal	Durable process history	Progress, approval, activity results
LangGraph	Bounded investigation loop	Reasoning and tools in an activity
Core Banking	Business records	Rules and payment idempotency

Event delivery, workflow progress and financial effects need separate controls.

Trace the durable business process



Model calls and network I/O run in activities, outside deterministic workflow code.

Stable identity closes two duplicate windows

BUSINESS CASE

case-501

The source identity stays stable across event redelivery.

WORKFLOW

dispute-case-case-501

A second event reuses the existing business process.

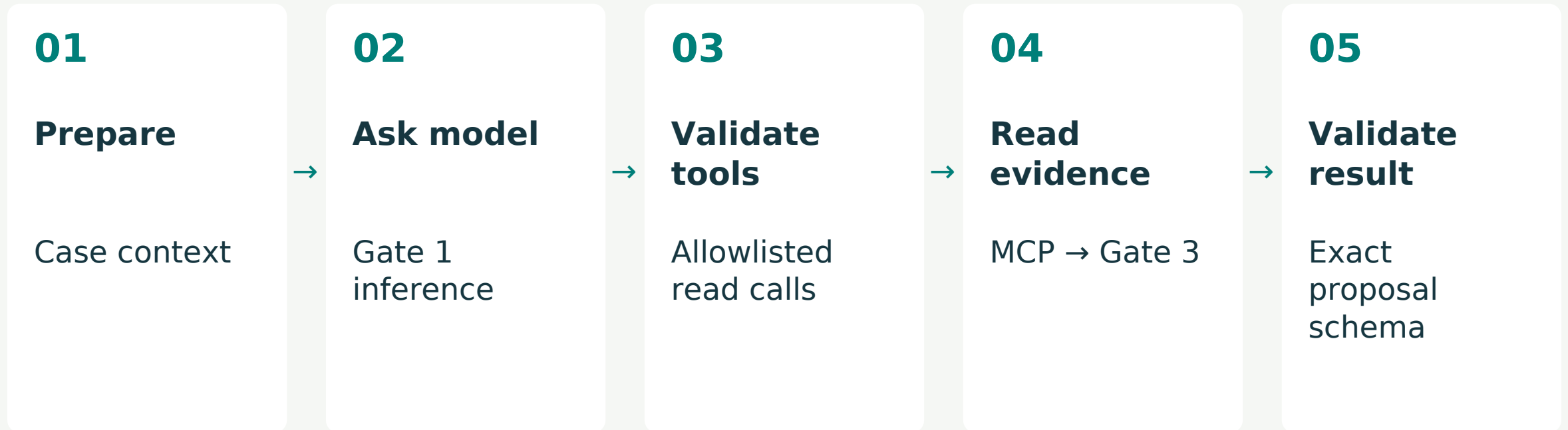
PAYMENT

settle-dispute-case-501

A settlement retry reuses the business effect.

Commit the Kafka offset after Temporal accepts or confirms the matching workflow.

Investigation has a bounded tool loop



The loop may repeat within its bounds. Diagnosis has no payment or approval tool.

Demo: deliver a case and inspect progress

Poll until the investigation reaches its approval wait.

TERMINAL / FROM REPOSITORY ROOT

```
.venv/bin/python workshops/w3/client.py emit \  
  --case-id case-501 --customer-id cust-8801  
  
.venv/bin/python workshops/w3/client.py query --case-id case-501
```

Inspect graph steps, selected tools, returned evidence and the actual proposal.

The agent proposes, then the process pauses

PHASE

WAITING_FOR_APPROVAL

PROPOSAL

Replay: ₹750 to acc-101. Record the actual amount and destination.

LEDGER

approval_decision:
null
No settlement
payment yet.

Server rules control the transition from proposal to business effect.

Exercise: capture the pre-crash checkpoint

Capture	Compare after recovery
Workflow ID	Same dispute-case-case-501
Proposal	Same amount, currency and destination
Current phase	WAITING_FOR_APPROVAL
Approval decision	Still pending
Settlement count	Still zero

Save the state before the failure so recovery becomes measurable.

Demo: stop, redeliver, restart

TERMINAL / FROM REPOSITORY ROOT

```
docker compose --profile w3 stop worker
```

```
.venv/bin/python workshops/w3/client.py emit \  
  --case-id case-501 --customer-id cust-8801
```

```
docker compose --profile w3 start worker
```

```
.venv/bin/python workshops/w3/client.py query --case-id case-501
```

The duplicate event can wait in Kafka while the worker is stopped.

Recovery preserves recorded progress

SURVIVES

Workflow identity, completed activity results, proposal and pending approval.

MAY REPEAT

A failed activity's inference and read tools can run again.

MUST NOT DUPLICATE

A financial settlement under the stable business idempotency key.

Durability does not mean every intermediate computation runs only once.

Review the recovered proposal, then approve

TERMINAL / FROM REPOSITORY ROOT

```
.venv/bin/python workshops/w3/client.py approve \  
  --case-id case-501 --reviewer ops-lead \  
  --comments "Reviewed case evidence and validated proposal"  
  
.venv/bin/python workshops/w3/client.py query --case-id case-501
```

The approval command waits for the workflow result.

Prove one approved settlement

Evidence	Expected fresh-run observation
Workflow phase	COMPLETED
Case status	resolved
Settlement key	settle-dispute-case-501
Payment count for that key	Exactly 1
Business amount	Matches the reviewed proposal

Reconcile workflow history with the actual banking record.

Three distinct failure windows

Failure	Control	Exercise scope
Event delivered twice	Stable workflow identity	Observed in redelivery
Worker dies during wait	Temporal durable state	Observed in restart
Payment commits; reply lost	Stable settlement key	Separate retry scenario

State the evidence you have for each recovery claim.

From a demo pause to a longer process

LAB

The approval timeout is 24 hours. The timed demo illustrates a durable wait.

PRODUCTION DESIGN

Choose timeout and escalation rules, persistence, availability and recovery procedures.

NEXT APPLICATIONS

Incident remediation and procurement can reuse the pattern with their own rules.

These extensions require design work beyond the implemented dispute lab.

Progress survives. Authority stays explicit.

Governed investigation + durable approval + idempotent settlement.

W4 / Test the whole system under an agent attack.

Presenter preparation & source guide

PREPARE

Activate w3. Use fresh local workflow state. Declare live/replay mode before the demo.

OPEN

Case, workflow query, worker logs, graph/tool evidence and payment records.

PRESERVE

Save previous evidence. The automated W3 rehearsal resets bank and Temporal lab state.

Use the manual exercise for delivery; reserve destructive rehearsals for disposable labs.